

ABCD4 (O14678) — Adjudication of Fatty-Acid & Peroxisome Function Claims

Focus: function_assignment · hypothesis `fatty-acid-functions-and-peroxisome-organization`
Source: `genes/human/ABCD4/ABCD4-ai-review.yaml` (free-text) **Date:** 2026-09-21 · Iteration 1

Executive Judgment

Verdict: Over-annotated / Refuted for all four claims (as direct ABCD4 functions).

The seed hypothesis proposes that human ABCD4 *retains* (i) fatty acid transmembrane transporter activity, (ii) fatty acid β -oxidation participation, (iii) very-long-chain fatty acid (VLCFA) biosynthesis participation, or (iv) peroxisome organization. Adjudicating each claim separately against its actual GO definition, the evidence codes on ABCD4, direct substrate/localization experiments, and the ABCD-subfamily phylogeny:

- **None of the four claims has any experimental support on ABCD4.** In EBI QuickGO (O14678, 55 annotations), every one of these terms is annotated **only by IBA** (Inferred from Biological Ancestor, ECO:0000318, GO_Central, GO_REF:0000033) — i.e. phylogenetically propagated from the ABCD1–3 ancestor, not from an ABCD4 experiment.
- **Claim 3 is factually mis-stated:** the *biosynthetic* term GO:0042761 is **not annotated to ABCD4 at all**; only the *catabolic* term GO:0042760 (also IBA-only) is present. ABC-D transporters import substrate for β -oxidation (catabolism), so a biosynthesis assignment is mechanistically backwards regardless.
- **Claim 4 (peroxisome organization) is contradicted by direct localization data:** ABCD4 localizes to the **ER and lysosome, not peroxisomes**, because it lacks the N-terminal peroxisomal-targeting region conserved in ABCD1–3

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- The experimentally grounded core function is lysosomal cobalamin (vitamin B12) export

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with MF term GO:0015420 supported by EXP/IDA/IMP.

The demonstrated cobalamin role does not, by itself, prove ABCD4 *cannot* carry a fatty-acyl substrate — but the burden of a positive claim is unmet: there is no direct ABCD4 fatty-acid transport/ β -oxidation assay, no peroxisomal residence, and the biosynthesis term is absent. **Recommended action: treat all four claims as non-core; do not assert them as ABCD4 functions.**

Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf limit
QuickGO O14678 (retrieved 2026-09-21)	Database (evidence codes)	Refutes (as direct claims)	1–4	All four seed terms are IBA-only ; GO:0042761 (VLCFA biosynthesis) not annotated at all	Human, curated GO	High prov IBA = exper
19010322 (Kashiwayama 2009)	Localization (IF, deletion constructs)	Refutes	4 (peroxisome)	P70R/ABCD4 localizes to ER, not peroxisomes ; lacks N-terminal hydrophilic peroxisomal- targeting region present in ABCD1–3	CHO cells, transfected	High over syste
27456980 (Kawaguchi 2016)	Localization + interaction	Refutes / qualifies	4	Endogenous ABCD4 is ER+lysosome; lysosomal delivery requires escort LMBD1 (lost on LMBRD1 KO)	Human cells	High endo assay
27766264 (Kawaguchi & Morita 2016)	Review synthesis	Supports (reframing)	1–4	ABCD1–3 = peroxisomal VLCFA/ branched acyl-CoA transporters; ABCD4 = lysosomal , vitamin B12 — explicit functional separation	Review	Revie well-
22922874 (Coelho 2012)	Mutant phenotype + localization	Refutes (legacy) / supports core	4 + core	cblJ mutations block lysosomal cobalamin release ; ABCD4 colocalizes LAMP1/LMBD1; gene "previously thought to have peroxisomal	Human patients/ fibroblasts	High landr prim

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf limit
				localization and function"		
33845046 (Reactome/ UniProt EXP)	Direct assay (EXP/IDA/ IMP)	Supports core	core MF	ABC-type vitamin B12 transporter activity (GO:0015420) experimentally supported	Human	High
31467407 (Xu 2019)	Structure (cryo-EM)	Supports core	core	Cryo-EM of human lysosomal cobalamin exporter ABCD4	Recombinant human protein	High
42303638 (Liu 2026)	Structure (cryo-EM)	Supports core	core / 4	ABCD4–LMBD1 complex; cobalamin recognition & transport cycle; LMBD1-dependent lysosomal trafficking	Recombinant human	High
23141461 (Kim 2012)	Mutant phenotype	Supports core	core	cblJ ABCD4 mutation blocks cobalamin transport lysosome→cytoplasm	Human patient	High
28572511 (Fettelschoss 2017)	Interaction/ mechanism	Supports core	core	ABCD4 ATPase/ clinical mutations disrupt cobalamin-pathway interaction	Human	Mod
9302272 / 14533738 (old)	Localization (IDA)	Competing/ legacy	4	Older IDA "peroxisome" CC annotations	Human	Low; super target studi
	Structural/ evolutionary	Qualifies/ supports	1–4	ABCD4 shares only ~34% identity with	Human paralogs,	Medi ident

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Conf limit
This report (computed, UniProt seqs)				ABCD1–3 (mean 33.8%) vs 49.2% among ABCD1/2/3; ABCD4 is shortest (606 aa), lacking the N-terminal peroxisomal-targeting extension	NW global alignment	≠sub align demo funct diver
33729671 / 23141461 / 22922874 (cblJ patients)	Mutant phenotype	Refutes (indirectly)	1–4	cblJ disease signature is combined methylnmalonic acidemia + homocystinuria (a cobalamin phenotype); no VLCFA accumulation / peroxisomal-disease phenotype reported	Human patients	Medi abser repor syste VLCFA

Provenance artifacts: /tmp/abcd4_claim_evidence_matrix.csv (QuickGO evidence codes) and the computed pairwise Needleman-Wunsch %identity table below (ABCD1–4).

Computed ABCD subfamily divergence (NW global alignment, match+1/mismatch–1/gap–1):

Pair	% identity
ABCD1–ABCD2	64.9
ABCD1–ABCD3	40.7
ABCD2–ABCD3	42.1
ABCD1–ABCD4	32.2
ABCD2–ABCD4	33.1
ABCD3–ABCD4	36.0
ABCD4 vs ABCD1-3 (mean)	33.8
ABCD1/2/3 among themselves (mean)	49.2

ABCD4 is the divergent out-group of the subfamily and the shortest member (606 aa vs 659–745 aa), consistent with the missing N-terminal peroxisomal-targeting region — the branch across which the IBA fatty-acid/peroxisome terms were propagated is exactly the weak, long branch where phylogenetic inference is least reliable.

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current evidence on ABCD4	Recommended action
GO:0005324 long-chain fatty acid transmembrane transporter activity ($\approx$ seed GO:0015245)	MF	IBA only	Do not assert as core; candidate for removal/NOT. No direct substrate assay; propagated from ABCD1–3.
GO:0006635 fatty acid β -oxidation	BP	IBA only	Non-core / candidate removal. No pathway participation shown for ABCD4.
GO:0042761 VLCFA biosynthetic process	BP	Not annotated	Claim unsupported; if a curator was tempted to add it, do not — mechanistically backwards and absent.
GO:0042760 VLCFA catabolic process	BP	IBA only	Non-core / candidate removal.
GO:0007031 peroxisome organization	BP	IBA only	Candidate removal; contradicted by ER/lysosomal localization.
GO:0005777 peroxisome / GO:0005778 peroxisomal membrane	CC	old IDA / IBA	Deprecate in favor of lysosome/ER; flag as superseded.
GO:0015420 ABC-type vitamin B12 transporter activity	MF	EXP/IDA/IMP	Retain as core MF.
GO:0015889 cobalamin transport; GO:0009235 cobalamin metabolic process	BP	IMP/IDA/TAS	Retain as core BP.
GO:0005765 lysosomal membrane	CC	EXP/IDA	Retain as core CC.

Not recommending "protein binding" (GO:0005515) as a substantive call — a specific, better-supported set of cobalamin terms is available.

Mechanistic Scope

- **Immediate molecular function tested:** ATP-driven transmembrane transport of a small-molecule substrate. ABCD4's demonstrated substrate is **cobalamin**, exported from the lysosomal lumen to the cytosol (handoff toward MMACHC), not a fatty acyl-CoA.
- **Direct activity vs. downstream:** The fatty-acid β -oxidation and VLCFA claims are *pathway-level* processes that in ABCD1–3 follow acyl-CoA import into peroxisomes. For ABCD4 there is neither the transport step (no assay) nor the compartment (not peroxisomal), so any such role would be inference, not direct activity.
- **Peroxisome organization** is an organelle-biogenesis process; ABCD4 is not a peroxisomal resident, removing the mechanistic basis for a direct role.

Conflicts and Alternatives

- **Paralog over-annotation (primary explanation):** ABCD4 shares the ABC-D half-transporter architecture with ABCD1 (X-ALD, VLCFA-CoA), ABCD2, and ABCD3 (branched acyl-CoA). GO's IBA pipeline propagates the ancestral fatty-acid/peroxisome functions to ABCD4. This is exactly the "frequency bias / paralog carry-over" the seed warns about.
- **Historical naming:** ABCD4 = "P70R / PMP69," named as a peroxisomal-membrane-protein relative, seeding legacy peroxisome CC annotations

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- **Seed's own term error:** the VLCFA claim names the *biosynthetic* process (GO:0042761), which is not annotated; only catabolic (GO:0042760) exists — reinforcing that the claim is loose paralog carry-over rather than a curated ABCD4 assignment.
- **Human loss-of-function phenotype is cobalamin-specific:** cblJ patients (ABCD4 biallelic mutations) present with combined **methylmalonic acidemia + homocystinuria**

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— the cobalamin signature — with **no reported VLCFA accumulation or peroxisomal-disease phenotype** (contrast X-linked ALD from ABCD1). This is indirect negative evidence against a physiologically important fatty-acid role, though not a systematic VLCFA measurement.

- **Sequence divergence (computed):** ABCD4 shares only ~34% identity with ABCD1–3 (vs 49% among the peroxisomal trio) and lacks their N-terminal targeting extension, so IBA transfer of ABCD1–3 attributes across this long branch is unreliable.
- **Not fully excluded:** no experiment has directly tested whether ABCD4 can bind/move a fatty-acyl species in vitro. Absence of evidence is not proof of substrate exclusivity — but it is insufficient to *assert* the function.

Knowledge Gaps

1. **Direct substrate specificity of ABCD4.** Checked: no reconstitution/transport assay with fatty-acyl-CoA substrates found. Matters because it would positively confirm or exclude a residual lipid-transport activity. Resolve with proteoliposome transport assays comparing cobalamin vs. (V)LCFA-CoA.
2. **Whether IBA annotations should be experimentally overridden.** Checked: GO_Central IBA present for all four. Matters for curation policy: IBA can be retained but should not outrank direct experimental CC/ME. A NOT-qualifier or removal request is the lead.
3. **Fate of legacy peroxisome CC annotations.** Checked: old IDA still present in QuickGO. Matters because CC drives downstream inference. Resolve by curator review citing

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Discriminating Tests

- **In vitro transport / ATPase stimulation** of purified ABCD4 ($\pm$ LMBD1) with cobalamin vs. C22:0/C24:0-CoA and branched acyl-CoA — the decisive substrate-specificity test.
- **Comparative localization** (endogenous, tagged CRISPR knock-in) vs. peroxisomal (PMP70/ABCD3) and lysosomal (LAMP1) markers — confirms non-peroxisomal residence.
- **VLCFA / plasma C26:0 profiling** in cblJ (ABCD4-deficient) patient fibroblasts vs. X-ALD (ABCD1) cells — if ABCD4 contributed to VLCFA handling, C26:0 should rise; the prediction under this report is **no VLCFA accumulation** in cblJ.
- **Phylogenetic branch-length / ancestral-state check** confirming the fatty-acid GO terms trace to the ABCD1–3 clade rather than a demonstrated ABCD4 attribute.

Curation Leads (require curator verification)

- **Action changes:** Mark GO:0005324, GO:0006635, GO:0042760, GO:0007031 (and any GO:0015245 if present) as **non-core / candidate removal or NOT** on ABCD4, noting they are IBA-only paralog propagation. Do **not** add GO:0042761 (VLCFA biosynthesis).
- **Retain as core:** GO:0015420 (MF), GO:0015889 + GO:0009235 (BP), GO:0005765 (CC).
- **Reconcile CC:** deprecate/annotate-superseded the legacy peroxisome CC terms in favor of lysosome/ER.
- **Candidate references + snippets to verify:**
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— "P70R-HA was localized to the endoplasmic reticulum (ER), not to peroxisomes."
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— "ABCD4 does not localize to peroxisomes but rather, the endoplasmic reticulum (ER), because it lacks the NH2-terminal hydrophilic region required for peroxisomal targeting."

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— "ABCD1-3 and ABCD4 are localized to peroxisomes and lysosomes, respectively."

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— "ABCD4, a gene that codes for an ABC transporter, which was previously thought to have peroxisomal localization and function."

- **Suggested questions for curator:** Are the four IBA fatty-acid/peroxisome terms worth retaining given definitive lysosomal/cobalamin evidence, or should they be removed/NOT-qualified to prevent downstream propagation?
- **Suggested experiments:** proteoliposome substrate panel; cblJ-fibroblast VLCFA profiling (see Discriminating Tests).

Limitations

IBA/evidence-code snapshot reflects QuickGO on 2026-09-21 and may differ from the review's frozen release. No wet-lab exclusion of fatty-acyl transport by ABCD4 exists; this report distinguishes *absence of positive evidence* from *proven exclusivity*.

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